

SAVANNA SUDAN CLIMATE

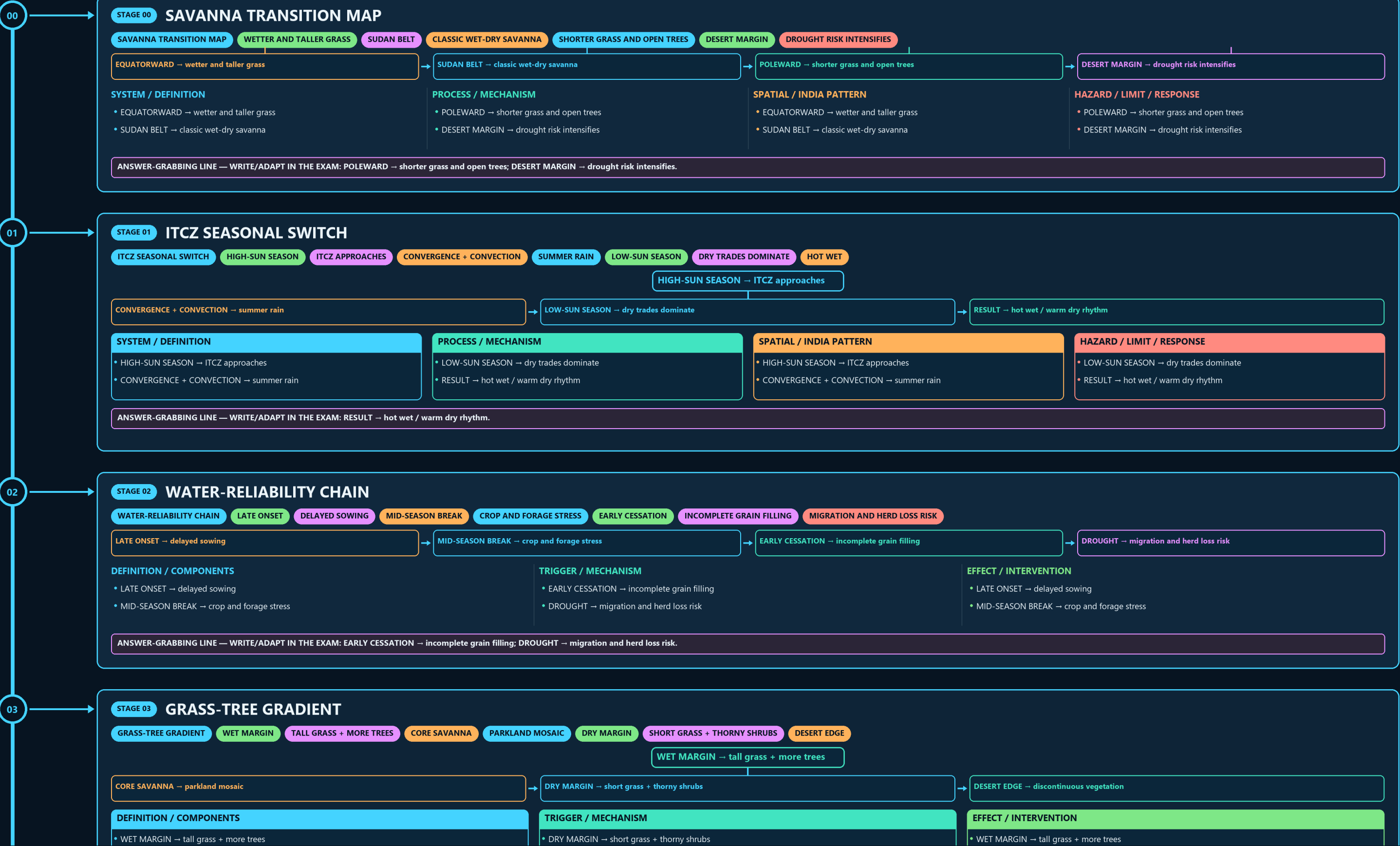
SAVANNA TRANSITION MAP → ITCZ SEASONAL SWITCH → WATER-RELIABILITY CHAIN → GRASS-TREE GRADIENT → FIRE-HERBIVORY BALANCE → WILDLIFE-LIVELIHOOD MATRIX

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARLY CESSATION → incomplete grain filling; DROUGHT → migration and herd loss risk.

WET MARGIN → tall grass + more trees

→ **DESERT EDGE** → discontinuous vegetation

EFFECT / INTERVENTION

- WET MARGIN → tall grass + more trees
- CORE SAVANNA → parkland mosaic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY MARGIN → short grass + thorny shrubs.

THICK BARK + ROOTS → selected tree survival

EFFECT / INTERVENTION

- SEASONAL RAIN → grass biomass
- DRY SEASON → combustible fuel

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THICK BARK + ROOTS → selected tree survival.

EXAM TRAP

PASTORALISTS → mobility spreads risk

CULTIVATORS → millet / sorghum timing

EXAM TRAP

- PASTORALISTS → mobility spreads risk
- CULTIVATORS → millet / sorghum timing

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PASTORALISTS → mobility spreads risk; CULTIVATORS → millet / sorghum timing.

➡ **FIXED WATER POINT** → local grazing concentration

EFFECT / INTERVENTION

- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY SEASON → fire and surface sealing; FIXED WATER POINT → local grazing concentration.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PASTORALISTS → mobility spreads risk; CULTIVATORS → millet / sorghum timing.

06

STAGE 06 DEGRADATION PATHWAY

DEGRADATION PATHWAY COVER LOSS RAINDROP AND WIND EXPOSURE WET SEASON RUNOFF, LEACHING AND EROSION DRY SEASON FIRE AND SURFACE SEALING FIXED WATER POINT

COVER LOSS → raindrop and wind exposure

WET SEASON → runoff, leaching and erosion

DRY SEASON → fire and surface sealing

FIXED WATER POINT → local grazing concentration

DEFINITION / COMPONENTS

- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

TRIGGER / MECHANISM

- DRY SEASON → fire and surface sealing
- FIXED WATER POINT → local grazing concentration

EFFECT / INTERVENTION

- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY SEASON → fire and surface sealing; FIXED WATER POINT → local grazing concentration.

07

STAGE 07 INDIA FOREST GRADIENT

INDIA FOREST GRADIENT MOIST DECIDUOUS LONGER WET SEASON DRY DECIDUOUS STRONGER LEAF-SHEDDING SEMI-ARID MOISTURE STRESS OVERLAPPING BELTS, NOT FIXED WALLS

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
MOIST DECIDUOUS → longer wet season	THORN / SCRUB → semi-arid moisture stress	MOIST DECIDUOUS → longer wet season	THORN / SCRUB → semi-arid moisture stress
DRY DECIDUOUS → stronger leaf-shedding	RULE → overlapping belts, not fixed walls	DRY DECIDUOUS → stronger leaf-shedding	RULE → overlapping belts, not fixed walls

SYSTEM / DEFINITION

- MOIST DECIDUOUS → longer wet season
- DRY DECIDUOUS → stronger leaf-shedding

PROCESS / MECHANISM

- THORN / SCRUB → semi-arid moisture stress
- RULE → overlapping belts, not fixed walls

SPATIAL / INDIA PATTERN

- MOIST DECIDUOUS → longer wet season
- DRY DECIDUOUS → stronger leaf-shedding

HAZARD / LIMIT / RESPONSE

- THORN / SCRUB → semi-arid moisture stress
- RULE → overlapping belts, not fixed walls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THORN / SCRUB → semi-arid moisture stress; RULE → overlapping belts, not fixed walls.

08

STAGE 08 INDIAN GRASSLAND FIREWALL

INDIAN GRASSLAND FIREWALL HUMID TALL GRASS AND FLOODPLAIN MONTANE GRASS-FOREST MOSAIC ARID-SALINE KACHCHH GRASSLAND HYDROLOGY AND HISTORY FIRST

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
TERAI → humid tall grass and floodplain	BANNI → arid-saline Kachchh grassland	TERAI → humid tall grass and floodplain	BANNI → arid-saline Kachchh grassland
SHOLA → montane grass-forest mosaic	MANAGEMENT → hydrology and history first	SHOLA → montane grass-forest mosaic	MANAGEMENT → hydrology and history first

SYSTEM / DEFINITION

- TERAI → humid tall grass and floodplain
- SHOLA → montane grass-forest mosaic

PROCESS / MECHANISM

- BANNI → arid-saline Kachchh grassland
- MANAGEMENT → hydrology and history first

SPATIAL / INDIA PATTERN

- TERAI → humid tall grass and floodplain
- SHOLA → montane grass-forest mosaic

HAZARD / LIMIT / RESPONSE

- BANNI → arid-saline Kachchh grassland
- MANAGEMENT → hydrology and history first

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BANNI → arid-saline Kachchh grassland; MANAGEMENT → hydrology and history first.

09

STAGE 09 RESTORATION DECISION TREE

RESTORATION DECISION TREE NATURAL OPEN ECOSYSTEM CONSERVE OPENNESS DEGRADED SCRUB RESTORE NATIVE MOSAIC WATER PROCESS RECHARGE AND DRAINAGE GRAZING AND TENURE DESIGN

NATURAL OPEN ECOSYSTEM → conserve openness

DEGRADED SCRUB → restore native mosaic

WATER PROCESS → recharge and drainage

LIVELIHOOD → grazing and tenure design

DEFINITION / COMPONENTS

- NATURAL OPEN ECOSYSTEM → conserve openness
- DEGRADED SCRUB → restore native mosaic

TRIGGER / MECHANISM

- WATER PROCESS → recharge and drainage
- LIVELIHOOD → grazing and tenure design

EFFECT / INTERVENTION

- NATURAL OPEN ECOSYSTEM → conserve openness
- DEGRADED SCRUB → restore native mosaic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER PROCESS → recharge and drainage; LIVELIHOOD → grazing and tenure design.

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EFFECT / INTERVENTION

- NATURAL OPEN ECOSYSTEM → conserve openness
- DEGRADED SCRUB → restore native mosaic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER PROCESS → recharge and drainage; LIVELIHOOD → grazing and tenure design.

10

STAGE 10 SAVANNA PYQ FIREWALL

SAVANNA PYQ FIREWALL 2021 PRELIMS TREE-LIMITING CONDITIONS LONG DRY SEASON FIRE + HERBIVORY NO ANSWER LETTER INVENTED

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
2021 PRELIMS → tree-limiting conditions	DISTURBANCE → fire + herbivory	2021 PRELIMS → tree-limiting conditions	DISTURBANCE → fire + herbivory
CLIMATE → long dry season	LEDGER → no answer letter invented	CLIMATE → long dry season	LEDGER → no answer letter invented
CLASSIFICATION 2021 PRELIMS → tree-limiting conditions - CLIMATE → long dry season	DIAGNOSTIC BASIS - DISTURBANCE → fire + herbivory - LEDGER → no answer letter invented	PROCESS LINK 2021 PRELIMS → tree-limiting conditions - CLIMATE → long dry season	EXAM TRAP - DISTURBANCE → fire + herbivory - LEDGER → no answer letter invented

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DISTURBANCE → fire + herbivory; LEDGER → no answer letter invented.

11

STAGE 11 SAVANNA ANSWER SPINE

SAVANNA ANSWER SPINE RAINFOREST-DESERT TRANSITION ITCZ AND WET-DRY SEASON GRASS-TREE DISTURBANCE BALANCE INDIA MOSAICS AND RESTORATION

LOCATE → rainforest-desert transition

DERIVE → ITCZ and wet-dry season

EXPLAIN → grass-tree disturbance balance

APPLY → India mosaics and restoration

DEFINITION / COMPONENTS

- LOCATE → rainforest-desert transition
- DERIVE → ITCZ and wet-dry season

TRIGGER / MECHANISM

- EXPLAIN → grass-tree disturbance balance
- APPLY → India mosaics and restoration

EFFECT / INTERVENTION

- LOCATE → rainforest-desert transition
- DERIVE → ITCZ and wet-dry season

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPLAIN → grass-tree disturbance balance; APPLY → India mosaics and restoration.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT INDIA'S SAVANNA ANALOGUE DRY DECIDUOUS FOREST + THORN SCRUB + GRASSLANDS . TROPICAL MOIST DECIDUOUS FORESTS 100–200 CM RAINFALL; DOMINANT TEAK AND SAL . MOIST DECIDUOUS REGIONS SAHYADRIS , NE PENINSULA, HIMALAYAN FOOTHILLS.

THORN/SCRUB REFLECTS ARIDITY PLUS LAND-USE PRESSURE; IT IS NOT

OPTIONAL PROCESS DEPTH

- India's savanna analogue = dry deciduous forest + thorn scrub + grasslands .
- Tropical moist deciduous forests: 100–200 cm rainfall; dominant teak and sal .
- Moist deciduous regions: Sahyadris , NE peninsula, Himalayan foothills.
- Thorn/scrub reflects aridity plus land-use pressure; it is not invariably degraded moist deciduous forest.

DATA / INSTITUTION LIMIT

- Rainfall thresholds overlap regionally; use approximate belts rather than rigid universal cut-offs.
- Thorn forest regions: peninsular India, Rajasthan, Haryana, Punjab, W UP, Kachchh, MP, Himalayan foothills.
- Deciduous trees shed leaves in dry season.
- Grasslands include Terai, shola and Banni examples in the source notes.

CONTEMPORARY PARALLEL

- Fire and grazing maintain open grassland/woodland mosaics.
- NOT: Dry deciduous forests are evergreen → They are deciduous and shed leaves in the dry season.
- NOT: Thorn forests occur in the wettest parts of India → They occur in arid/semi-arid belts, generally
- NOT: Banni and Terai grasslands are the same type → Terai is humid tall wet grassland; Banni is arid/saline Kachchh grassland.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.